Multimedia Wireless Networks A2

ns-3 Setup & First Scenario

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Basic Path

Goal



Does EDCA prioritization still work with realistic multimedia traffic?

ns-3 setup

```
❯ ./ns3 build
[0/2] Re-checking globbed directories...
[1885/2635] Building CXX object contrib/netsimulyzer/examples/CMakeFiles/wifi-bianchi-netsimulyzer.dir/wifi-bianchi-net
simulyzer.cc.o
/home/sascha/ns-allinone-3.48/ns-3.48/contrib/netsimulyzer/examples/wifi-bianchi-netsimulyzer.cc: In function ‘int main
(int, char**)’:
/home/sascha/ns-allinone-3.48/ns-3.48/contrib/netsimulyzer/examples/wifi-bianchi-netsimulyzer.cc:3417:22: warning: vari
able ‘k’ set but not used [-Wunused-but-set-variable=]
 3417 |             uint32_t k = 0;
      |                      ^
[2634/2635] Linking CXX executable /home/sascha/ns-allinone-3.48/ns-3.48/build/utils/ns3.48-test-runner-default
Finished executing the following commands:
/usr/bin/cmake --build /home/sascha/ns-allinone-3.48/ns-3.48/cmake-cache -j 7
```

Item	Value
ns-3 version	ns-3.48
Operating system	Fedora Linux 44
Kernel	7.2.7-200.fc44.x86_64 (64 bit)
Desktop	KDE Plasma 6.7.5 (Frameworks 6.30.0, Qt 6.11.2), Wayland
Device	Lenovo ThinkPad X1 Carbon Gen 9 (20XXS33900)
CPU	Intel Core i7-1165G7 @ 2.80 GHz, 8 threads
Memory	32 GiB (31.1 GiB usable)
GPU	Intel Iris Xe Graphics

```
❯ ./test.py
[0/2] Re-checking globbed directories...
ninja: no work to do.
Finished executing the following commands:
/usr/bin/cmake --build /home/sascha/ns-allinone-3.48/ns-3.48/cmake-cache -j 7
```

```
989 of 989 tests passed (989 passed, 0 skipped, 0 failed, 0 crashed, 0 valgrind errors)
```

3GPP TR 26.926 V19.0.0 (Rel. 19)

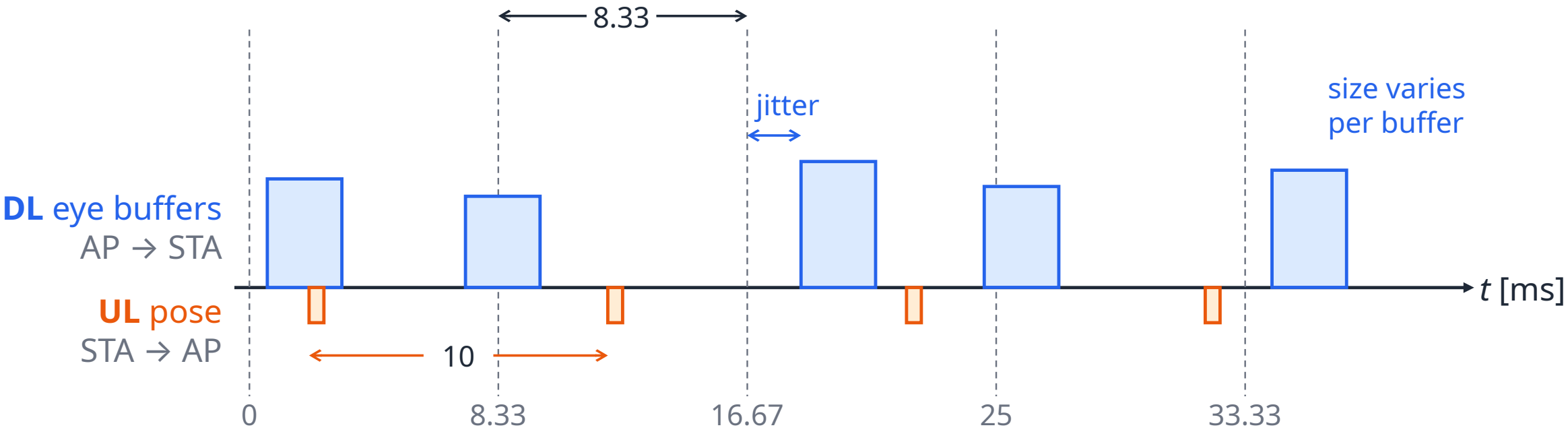
TS
Technical Specification

TR
Technical Report

Table 6.5.3.1-1: Statistical parameter values for dual eye buffer frame size (TR 38.838)

Parameter	unit	values for evaluation	Optional values for evaluation
Mean: M	byte	R×1e6 / (2×F) /8	R×1e6 / (2×F) / 8
STD	byte	10.5% of M	4% of M
Max	byte	150% of M	112% of M
Min	byte	50% of M	88% of M
R: data rate of the flow in Mbps F: frame generation rate of the flow in fps			

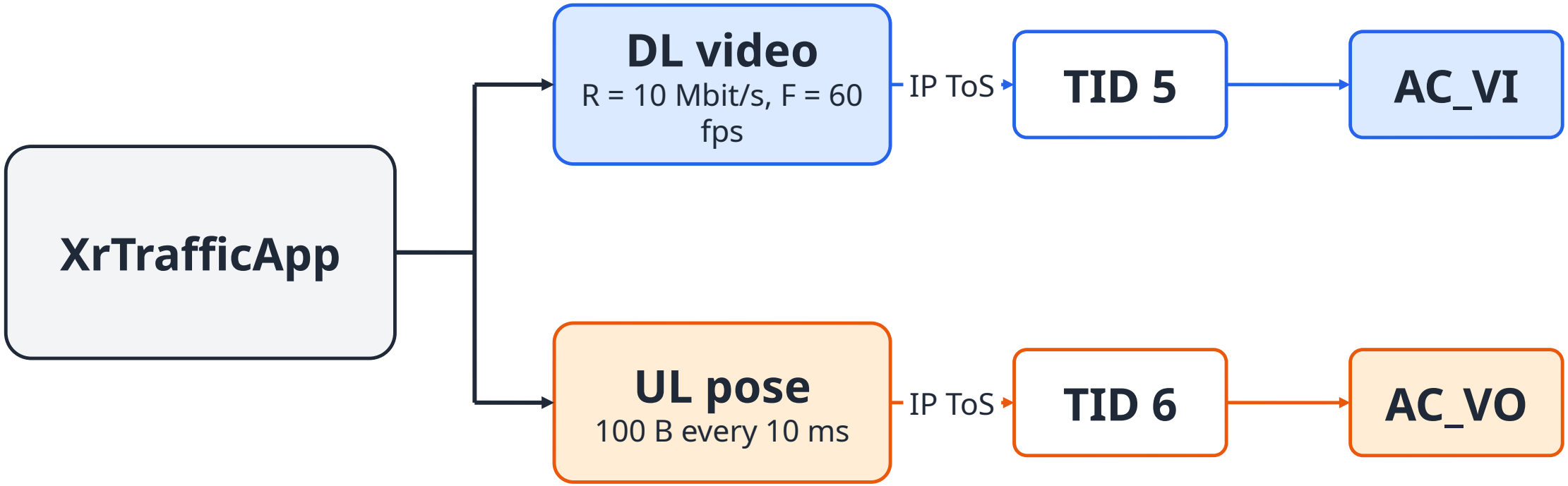
3GPP Traffic Model



Frame size: truncated Gaussian, $M = R \cdot 10^6 / (2F) / 8$, STD 10.5 % of M, ±50 %
Jitter: truncated Gaussian, STD 2 ms, ±4 ms

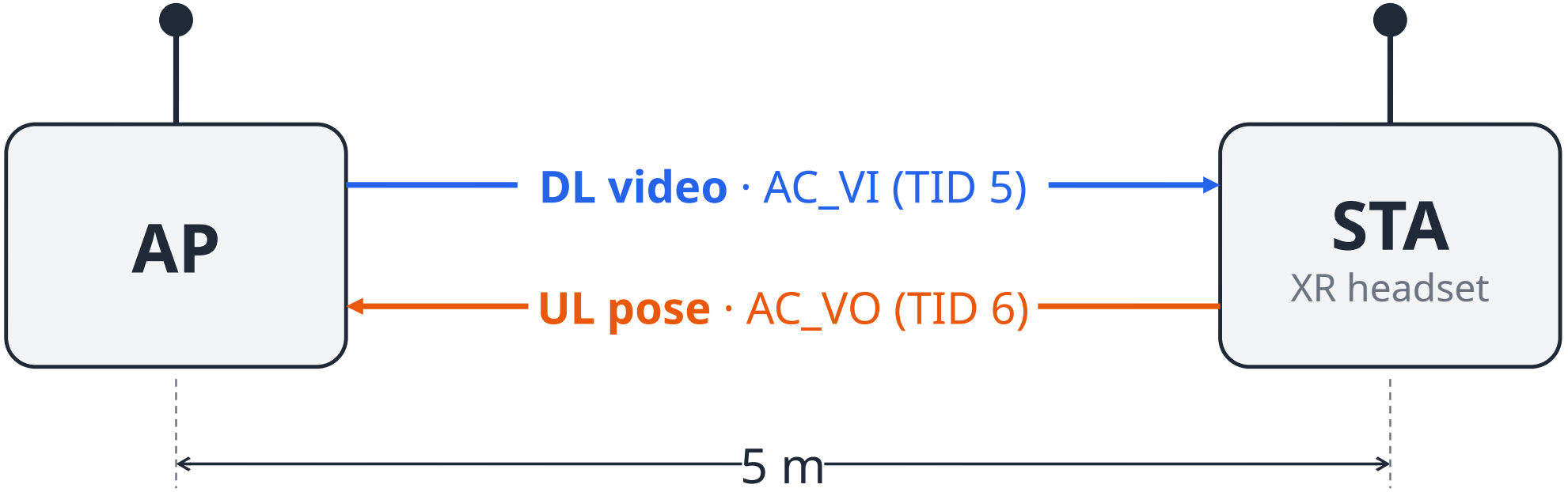
Based on Table 6.5.3.1-1

Traffic Generator



ToS byte → 802.11 user priority (TID) → access category

Scenario



802.11a · 54 Mbit/s · channel 36

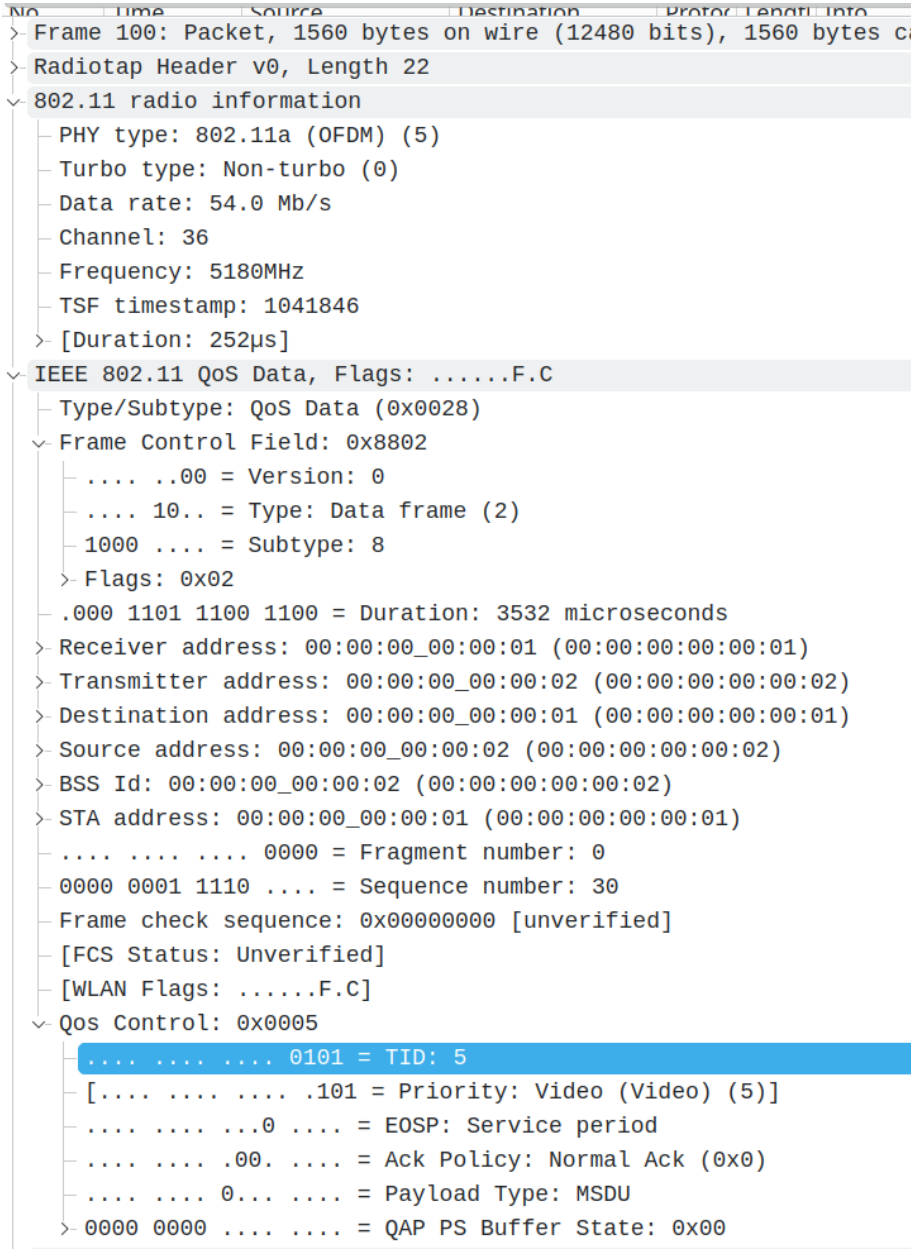
Results

	DL video	UL pose
throughput	10.2 Mbit/s	0.07 Mbit/s
mean delay	1.29 ms	0.37 ms
loss	0	0

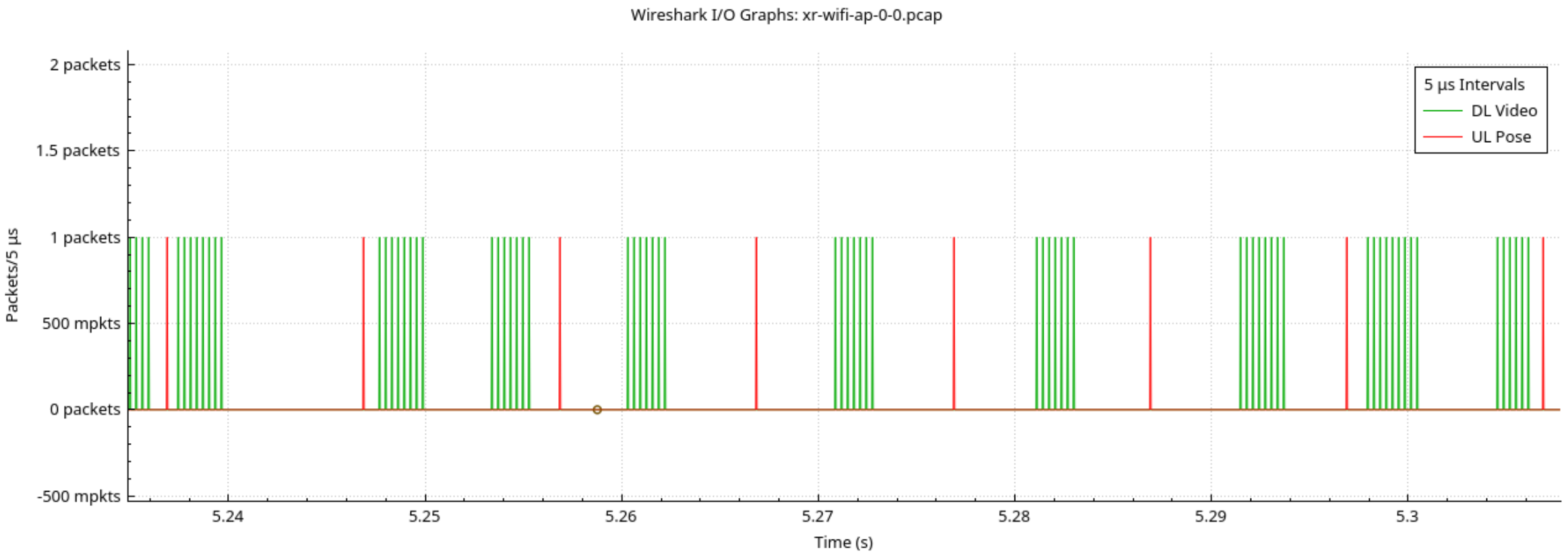
overhead DL video: 7.58 packets/frame × 120 frames/s ≈ 909 packets/s

909 × 28 B × 8 ≈ 0.20 Mbit/s

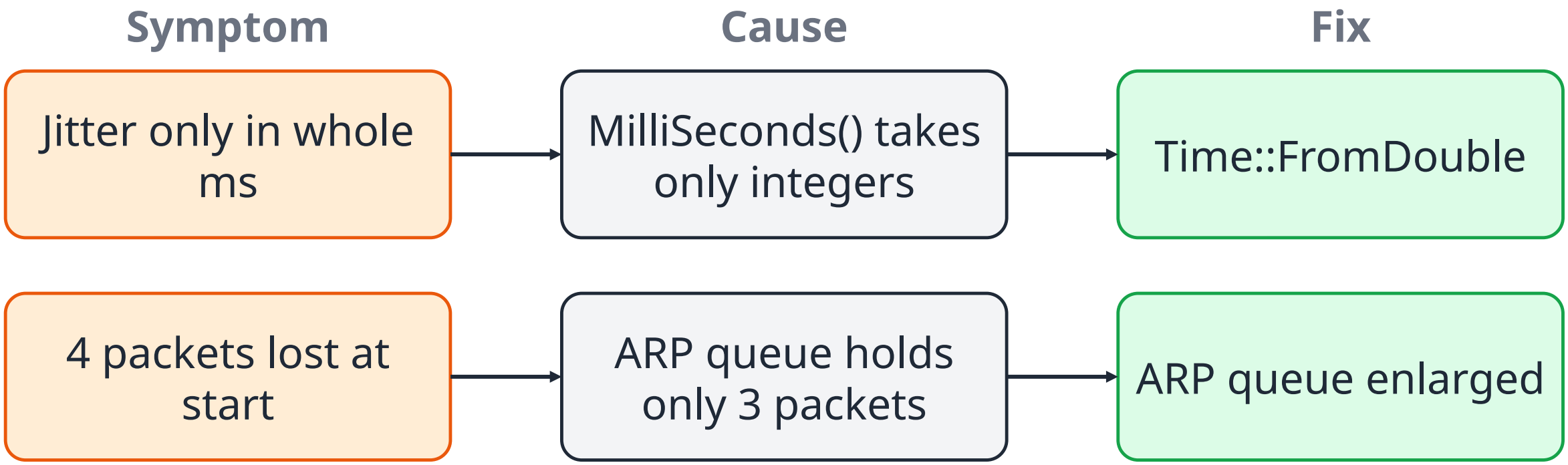
throughput UL pose: 100 packets/s × 88 B × 8 bit = 70 400 bit/s ≈ 0.070 Mbit/s



Results


Wireshark I/O Graphs: xr-wifi-ap-0-0.pcap
2 packets
1.5 packets
1 packets
500 mpkts
0 packets
-500 mpkts
Packets/5 µs
5.24
5.25
5.26
5.27
5.28
5.29
5.3
Time (s)
5 µs Intervals
DL Video
UL Pose

What broke


Symptom
Cause
Fix
Jitter only in whole ms
MilliSeconds() takes only integers
Time::FromDouble
4 packets lost at start
ARP queue holds only 3 packets
ARP queue enlarged

Recap

